Nihiro Tamura

+81 90-9286-0216 | tamura.nihiro@irl.sys.es.osaka-u.ac.jp | linkedin.com/in/nihiro-tamura

Education

- **The University of Osaka** — Osaka, Japan
  Ph.D. student in Engineering Science — *April 2026 – PRESENT*
- **The University of Osaka** — Osaka, Japan
  Master's degree in Engineering Science — *April 2024 – March 2026*
- **Kyoto Institute of Technology** — Kyoto, Japan
  Bachelor's degree in Mechanical Engineering — *April 2020 – March 2024*
  - **Graduated at the top of the program and received the Academic Excellence Award.**

Experience

- Research Assistant / Doctoral Researcher — Apr. 2026 – Present
  The University of Osaka, Ishiguro Lab — *Osaka, Japan*
  - **Develop an Android-specific control system for stable motion control.**
  - Implement feedforward control in addition to active disturbance rejection control (ADRC) to improve stability in difficult motion positions.
  - Generate feedforward control inputs via optimal control to achieve stable and precise motion.
- Research Assistant / Master's Researcher — Apr. 2024 – Mar. 2026
  The University of Osaka, Ishiguro Lab — *Osaka, Japan*
  - **Proposed a quantitative approach to evaluating the human-like motion of Androids.**
  - Developed an engineering metric that can quantitatively evaluate the human-likeness of movements realized by an arm robot with the same mechanism as an android arm.
  - As the first author, led the design, implementation, and experimental evaluation of the proposed metric.
  - **Published the research as a peer-reviewed international journal article:**
    N. Tamura, S. Ikemoto, T. Uchida, and H. Ishiguro, "Quantification of Human-Like Motion of Androids in Relation to Control Parameters," *IEEE Access*, vol. 14, pp. 17196–17205, 2026, doi: 10.1109/ACCESS.2026.3659342.
- Undergraduate Researcher — Dec. 2022 – Mar. 2024
  Kyoto Institute of Technology, Robotics Lab — *Kyoto, Japan*
  - **Developed a dynamic mathematical model of a vacuum-driven soft actuator for simulation and feedback control design.**
  - Identified model parameters from measured pressure-angle characteristics and step responses.
  - Validated the model through simulations and demonstrated the feasibility of angle control using PID control.

Projects

- **ROS 2-Integrated Optimal Control Framework** | *Python, ROS 2, Linux, WSL, Git* — 2026 – Present
  - Developed an optimal control computation framework based on the Euler–Lagrange method.
  - Implemented ROS 2 communication to send control parameters and subscribe to real-time measurement data.
  - Generated optimized feedforward control inputs from measured data and transmitted them to the robot via ROS 2.
- **ROS 2–Integrated Embedded Control System** | *Arduino, ROS 2, Git* — 2024 – 2026
  - Developed embedded control programs compatible with ROS 2 using Arduino IDE.
  - Implemented control algorithms including PID control and active disturbance rejection control (ADRC).
  - Integrated real-time communication between embedded controllers and ROS 2 for robotic control.
- **Soft Actuator Design and Numerical Simulation** | *Autodesk Inventor, MATLAB* — 2022 – 2024
  - Designed and developed a vacuum-driven soft actuator using Autodesk Inventor.
  - Developed a mathematical model of the soft actuator in MATLAB based on its mechanical structure.
  - Performed numerical simulations to analyze the actuator dynamics and control performance.

Skills

- Programming: **Python, C/C++, Arduino**
- Tools: **ROS 2, MATLAB, Linux, WSL, CAD**, KiCAD, Git, Docker, VS Code, PyTorch
- Languages: Japanese, English